FSE Training

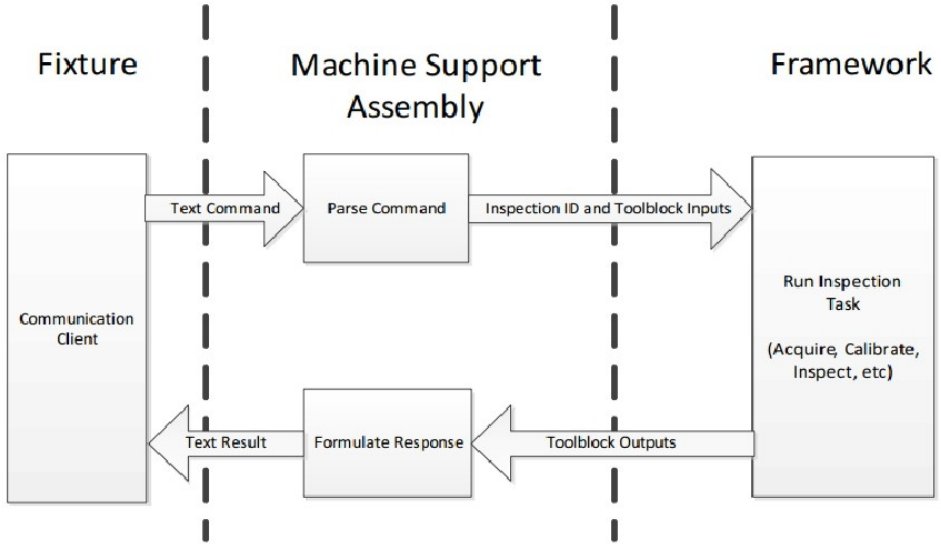
Content Sharing Introduction

- VS2012 界面讲解
- Machine Support 框架介绍
- Machine Support 的简单调试
- VPP 之间数据流向
- 3D 相机相关问题讲解

Universal Framework + Machine Support

➢ 运行流程

- AssemblyPlus 框架的运行流程如下图所示， Fixture 表示机器部分，其由 SI 控制， Framework 相当于一个视觉服务器，对视觉部分进行处理计算 (例如运行 Inspection, 光源配置等），而 Machine Support 则作为 Fixture 和 Framework 的中间桥梁负责命令的解析和响应，解析 Fixture 传来的指令，然后确定 Framework 中某个 Inspection 的运行，再将处理得到的结果进行打包反馈给 Fixture 。



MachineSupport 作用

Cognex.VisionPro.Comm.xml	2018-02-08 3:57	XML 文档
Cognex.VisionPro.Vs.TexturedRangeI...	2019-04-13 11:10	DLL 文件
Cognex.VPE.Core.dll	2020-04-07 7:22	DLL 文件
Cognex.VPE.ImageGenerator.Control...	2020-04-07 10:26	DLL 文件
Cognex.VPE.InspectPro.CVL.dll	2020-04-07 7:24	DLL 文件
Cognex.VS.AlignPlusUtilities.dll	2019-04-13 11:10	DLL 文件
Cognex.VS.Core.dll	2020-04-07 7:19	DLL 文件
Cognex.VS.FrameworkCommunicatio...	2020-09-15 14:31	DLL 文件
Cognex.VS.FrameworkSetup.dll	2020-10-28 20:20	DLL 文件
Cognex.VS.Interop.dll	2020-04-07 7:17	DLL 文件
Cognex.VS.LightController.dll	2020-09-15 14:31	DLL 文件
Cognex.VS.MachineSupport.dll	2021-12-10 16:43	DLL 文件
Cognex.VS.MachineSupport.pdb	2021-12-10 16:43	Program I
Cognex.VS.ToolBlockUtility.dll	2020-09-15 14:31	DLL 文件
Cognex.VS.Utility.dll	2020-09-15 14:31	DLL 文件
Cognex.VS.Utility.xml	2020-09-15 14:31	XML 文档
CognexAssemblyPlus.exe	2020-10-28 20:20	应用程序
CommunicationTester.exe	2018-02-22 6:51	应用程序
config.xml	2021-06-11 20:56	XML 文档
Configuration_General.ini	2021-12-10 16:45	配置设置
Configuration_NonGeneral.ini	2021-12-09 19:07	配置设置

运行流程：

- 接收并解析指令
- 给指定 Inspection 输入参数赋值
- 运行指定 Inspection
- 获取运算结果
- 封装并发送指令

其他功能：定义设备名称、定义 Simulator 、数据保存、数据传递、坐标计算…

Machine Support 结构

返回设备名称

返回模拟器数量

返回模拟器指令表（String）

返回模拟器 PLC 指令表（16 位数字）

```
public string MachineName...
public List<int> SimulatorCommandParserIDs...
public List<string> SimulatorCommandStrings...
public List<Tuple<String, Int16>> SimulatorPLCCommands...

public void ForceRecordFlush()...
public Cognex.VisionPro.CogGraphicCollection GetAcquisitionSetupOverlayGraphics(int
public void Init(IFrameworkSupport framework)...
public ProfinetUserData ProfinetJobParser(CommandAndInfo command)...
public void Shutdown()...
public string TCPAcqArmed(CommandAndInfo command)...
public string TCPAcqComplete(CommandAndInfo command)...
public string TCPJobParser(CommandAndInfo command)...
```

清缓存

实时模式添加 Graphic

初始化

处理 Profinet 接口

关闭

收到指令被调用

取像完成被调用

取像前被调用

Machine Support Key function

Init()

a) Framework Backup （框架备份）
b) Read Machine support Setting
c) Framework Custom Menu Setting
d) Hardware Prepare （硬件准备）

```
public void Init()
  {
    //allocate resourece
  }
```

Shutdown()

Shutdown is called once when the framework is closing. Use this function to cleanup anything initialized explicitly in the Init function. Most machines can leave this implementation empty.
（当框架关闭时调用 Shutdown 。使用此函数可清除在 Init 函数中显式初始化的任何内容。大多数机器可以将此实现留空。）

```
public void Shutdown()
{
  // If no resources were
allocated, there is nothing to clean up
}
```

Machine Support Key function

```
        }
        //Define RunJobParameters and add in the List
        ulong partID = ImageSaveQueue.gOnly.NewPart(CommandData.SerialNumber);
        RunJobParameters rjParams = new RunJobParameters(CommandData.InspectionID, partID);
        rjParams.Inputs.Add("I_Method", methodID);
        rjParams.Inputs.Add("InputCommand", command.Command);
        rjParams.Inputs.Add("ImageID", command.Command.Substring(0,13) + CommandData.SerialNumber);
        vrjParams.Add(rjParams);
        vCMDDatas.Add(CommandData);
        //DisplayGraphic(CommandData.PoseID, CommandData.NozzleID, 1);

    }
    //RunJob
    bool success = m_framework.RunJob(command, LightControlAction.AutoOnAndOff, true, vrjParams.ToArray());

    for (int i = 0; i < vrjParams.Count; i++)
    {
        errorCode = (int)vrjParams.ToArray()[i].Outputs["ErrorCode"];
```

Machine Support Key function

```
bool RunJob( CommandAndInfo command, LightControlAction lightControlAction,
      bool shareLightingSettings, params RunJobParameters[] parameters );
```

RunJob 是 IFrameworkSupport 接口中重要的方法。 RunJob 同时运行一个或多个检查 job，使用一组 RunJobParameters 对象来指定要运行的检查、存储输入、和 toolblocks.RunJob 的输出应该总是从 IMachineSupport 的主体内部调用 TCPJobParser() 函数。 每次调用 TCPJobParser 只能调用一次。每个 CommandAndInfo 和 RunJobParameters 对象只能用作 RunJob ONCE 的参数。

RunJob 的 4 个参数 .

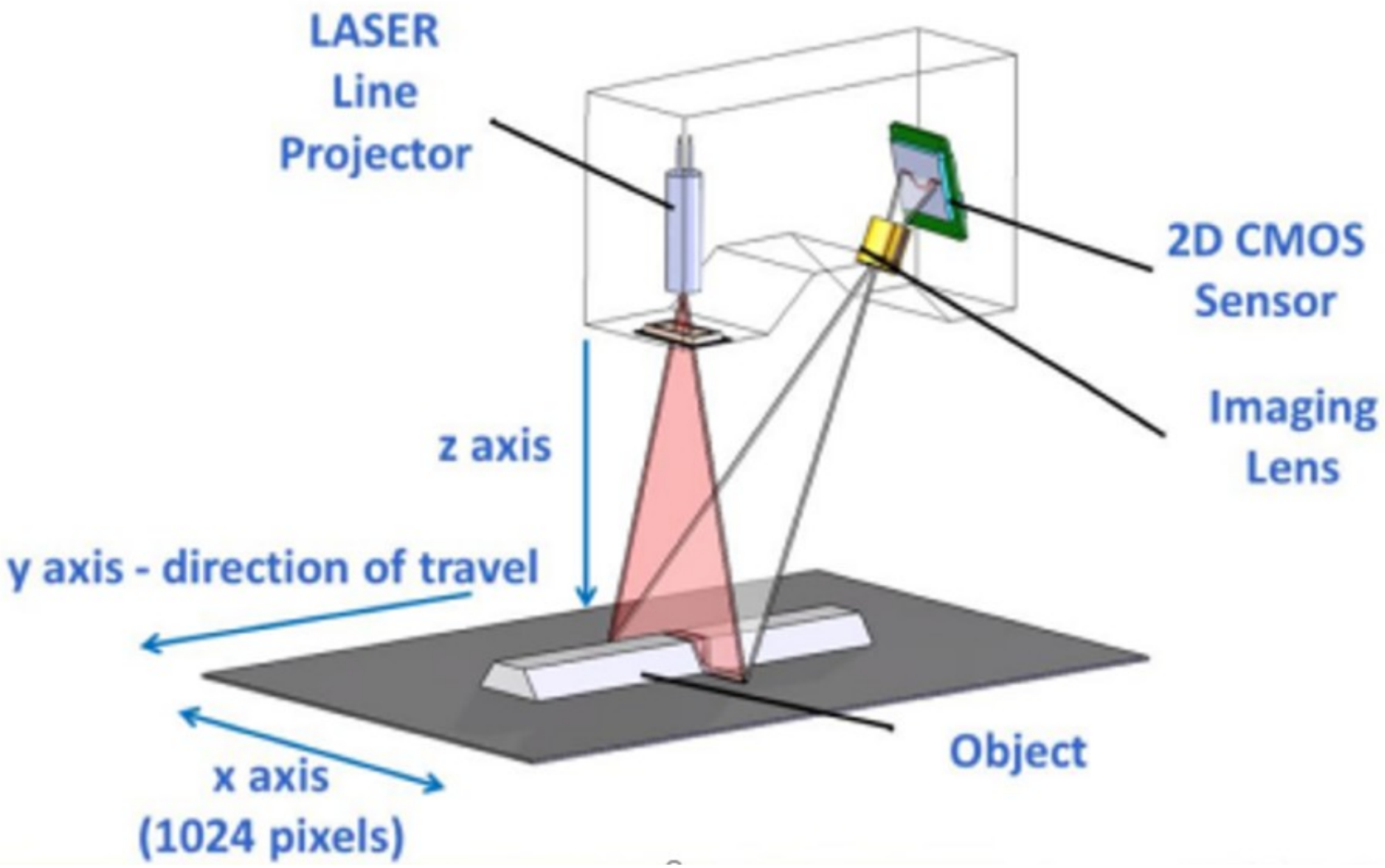
- 第一个是一个 CommandAndInfo 对象， command 。 这不应该被构造 .
- 第二个是 LightControlAction 。 这几乎总是 LightControlAction.AutoOnAndOff 或 LightControlAction.NoAction ，这取决于框架是否应该管理照明。
- 第三个参数是布尔值， shareLightingSettings 。 RunJob 一次可以运行多个检测工具，每个检测工具可以有自己的光照设置。如果 shareLightingSettings 为 true ，则第一次检查的照明设置将用于所有检查。如果 shareLightingSettings 为 false ，则每次检查都将进行轮流获取图像并使用单独的照明设置。这比共享照明设置慢得多
- 第四个参数是 RunJobParameters 对象的集合。 每个 RunJobParameters 对象代表一次单独检查的运行。

Machine Support Key function

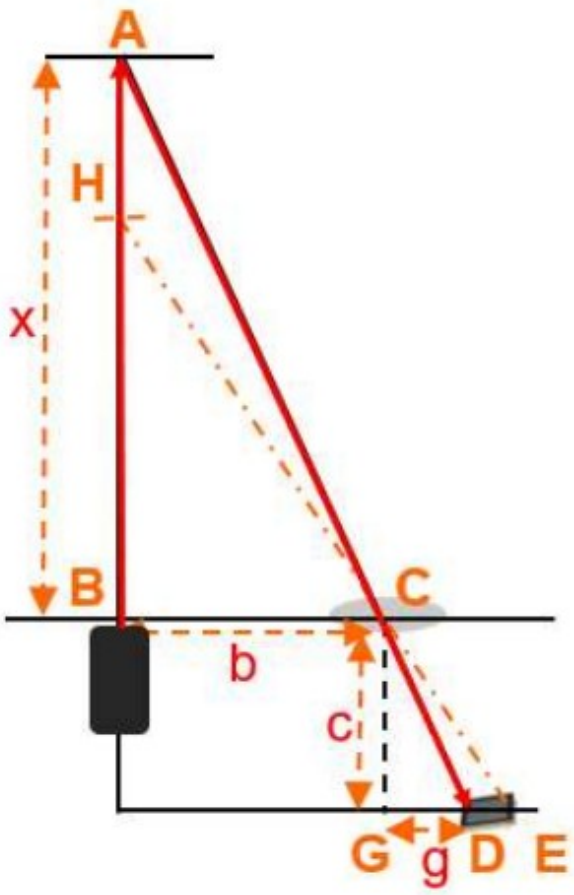
用 MessageManager 类向 Log 中增加信息

```
catch
{
  MessageManager.gOnly.Alarm("Image Group Failed.");
}
```

DS Camera


LASER
Line
Projector
2D CMOS
Sensor
Imaging
Lens
z axis
y axis - direction of travel
Object
x axis
(1024 pixels)

DS Camera



已知:
三角形ABC和CGD，其中
CG = c,
GD = g,
BC = b,
求**AB**的长度?

解：
根据相似三角形原理可知，设
AB = x，得：

$$\frac{x}{c} = \frac{b}{g} \quad => x = \frac{b*c}{g}$$

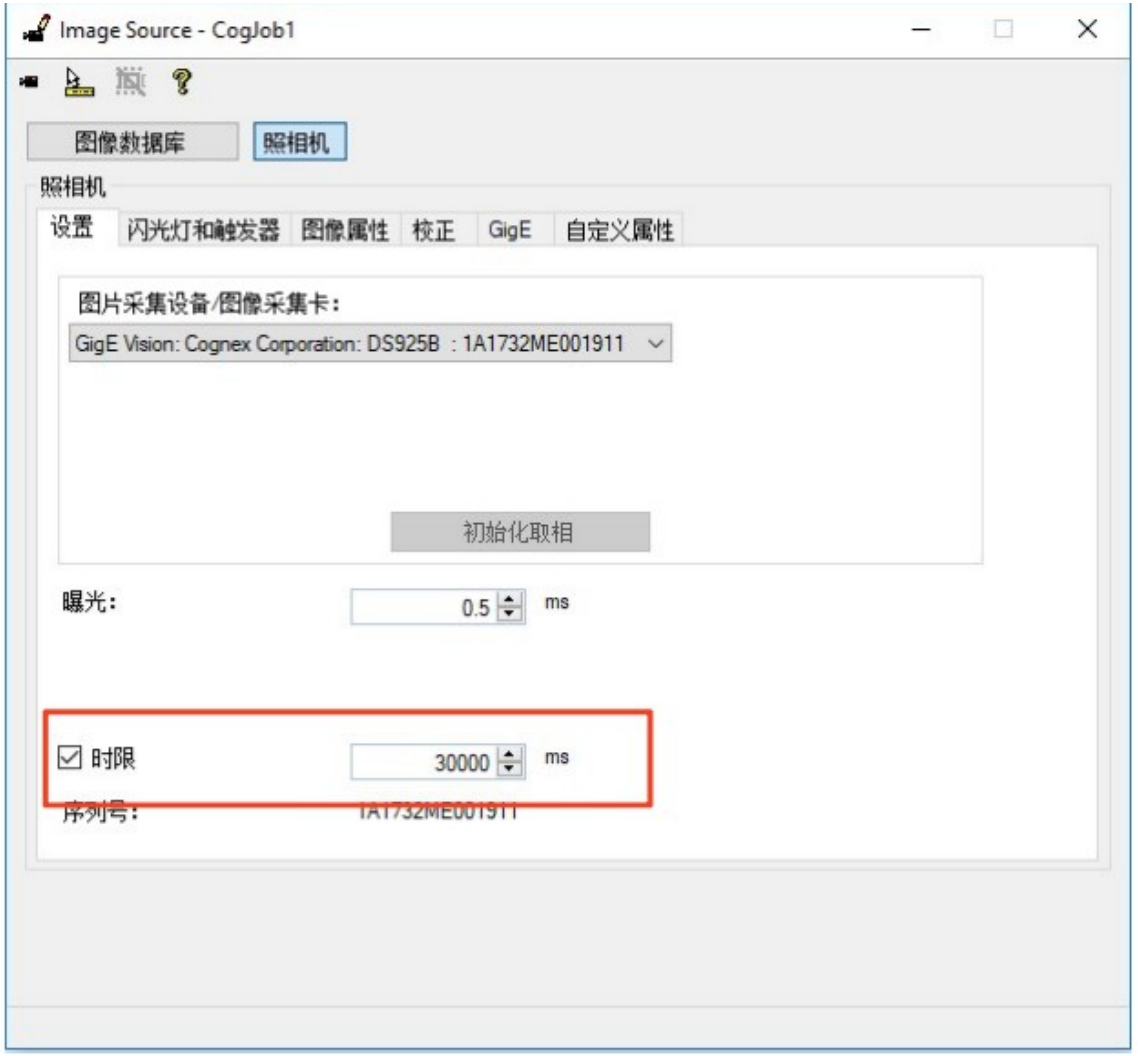
三角测量法

DS Camera

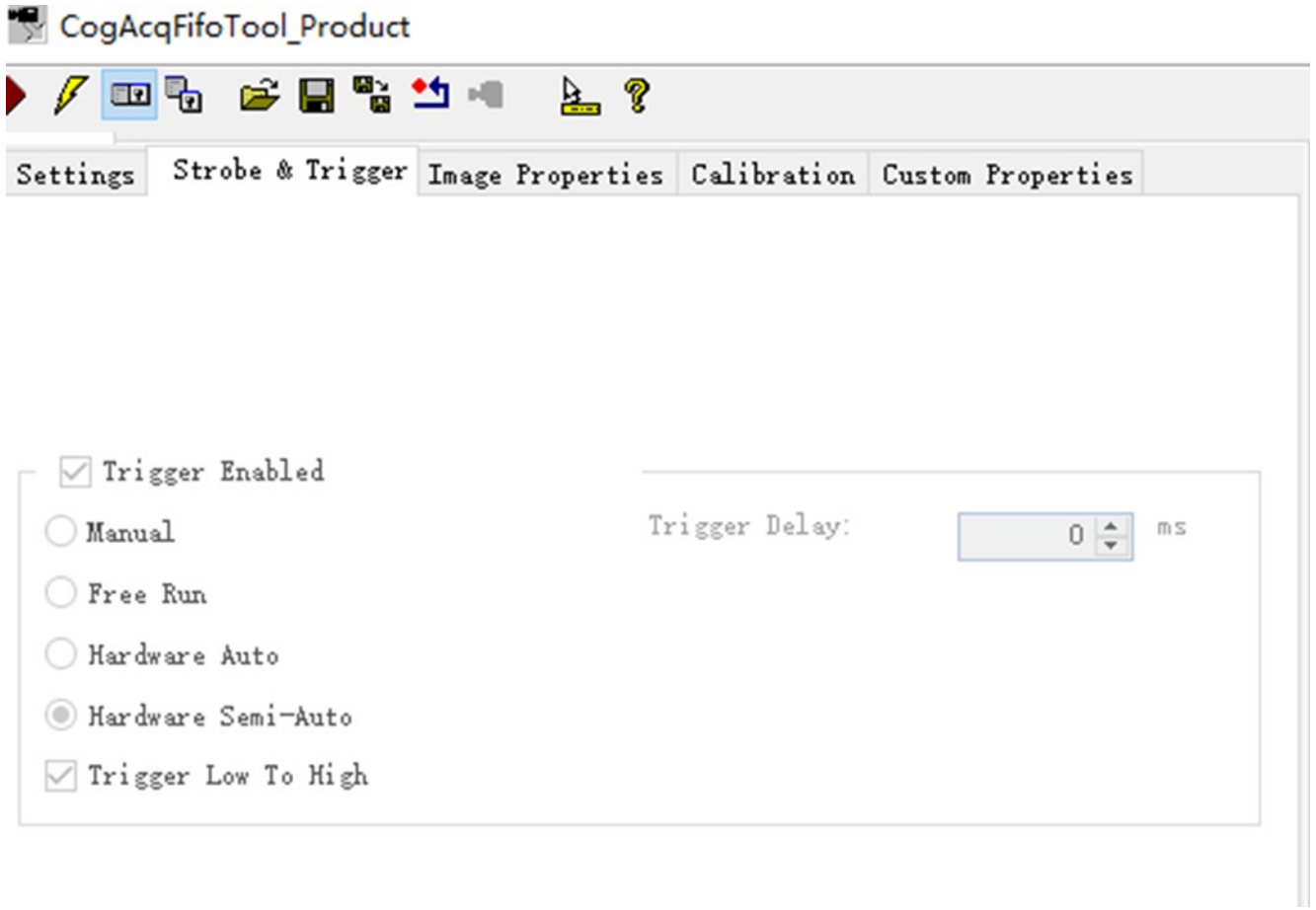
设置适当的超时，默认（30000ms）

如果启用，则此设置（以ms为单位）应设置为大于以下三个因子的总和：

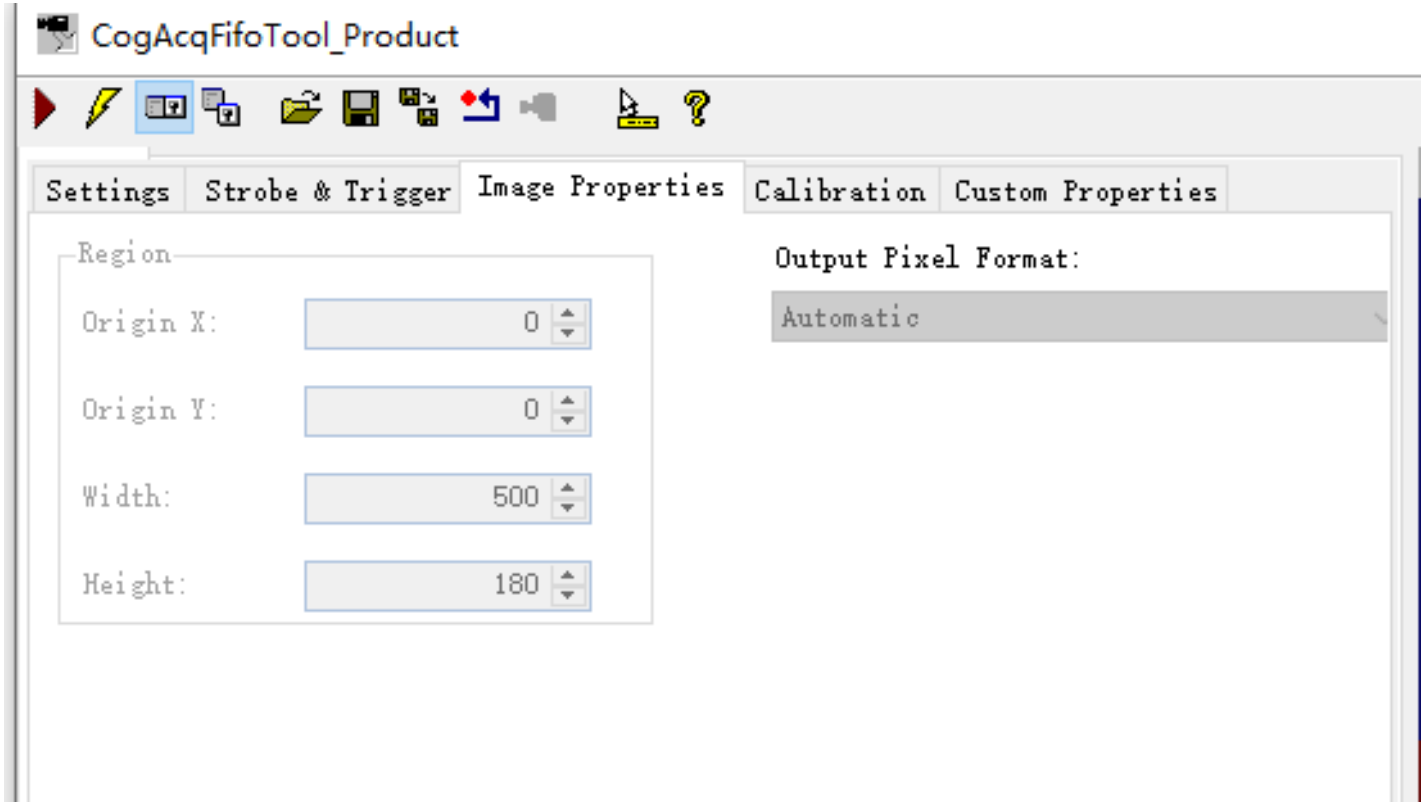
- 采集请求和采集开始之间的最大时间（可能是由于使用编码器时启动运动的延迟）
- 扫描零件所需的最长时间（由运动速度和扫描长度控制）；
- 扫描完成后完成图像数据传输所需的最长时间。



DS Camera

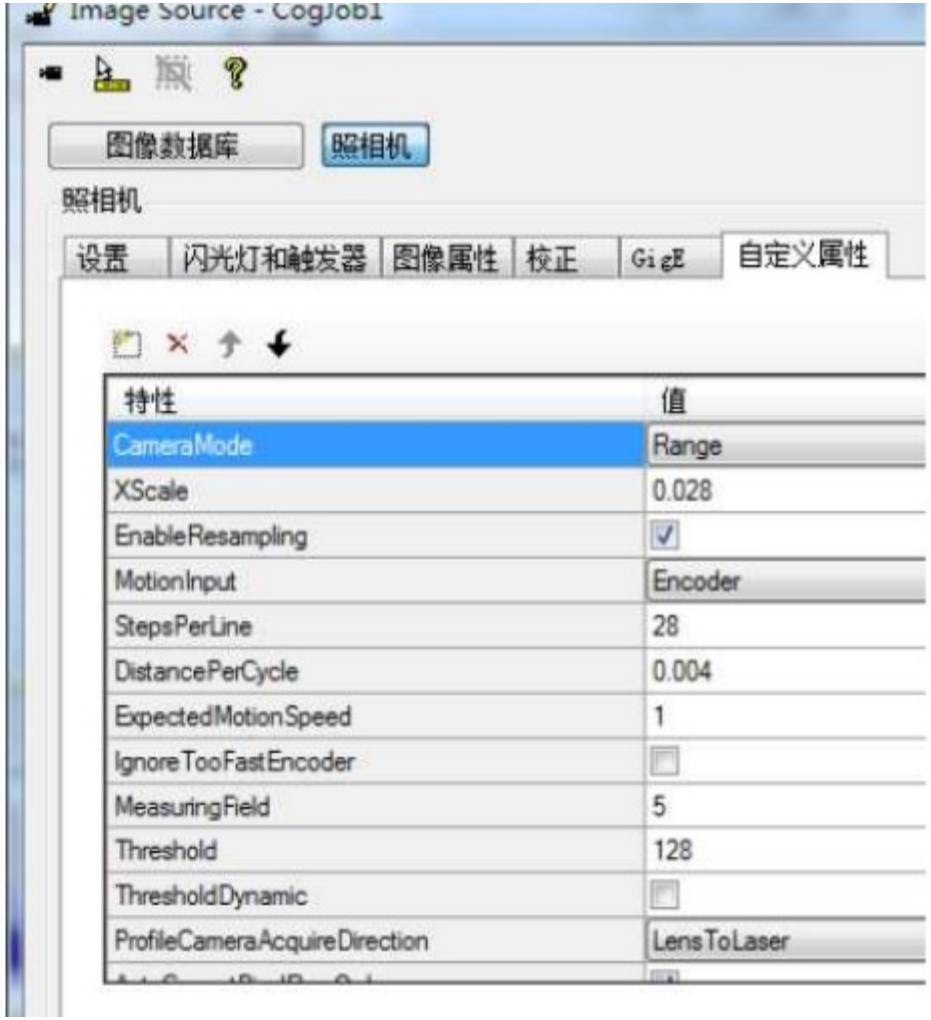

CogAcqFifoTool_Product
Settings
Strobe & Trigger
Image Properties
Calibration
Custom Properties
Trigger Enabled
Manual
Free Run
Hardware Auto
Hardware Semi-Auto
Trigger Low To High
Trigger Delay:
0
ms

DS Camera


CogAcqFifoTool_Product
Settings
Strobe & Trigger
Image Properties
Calibration
Custom Properties
Region
Origin X:
0
Origin Y:
0
Width:
500
Height:
180
Output Pixel Format:
Automatic

DS Camera

- X-Scale: 在X方向的像素分辨率
- StepsPerLine：采集一个像素需要的脉冲数
- Distance PerCicle： 一个周期内（四个脉冲信号）．平台移动的距离
- Measure Filed： 相机在Intensity模式下，的一个视野大小．不同的视野对应着不同的采样频率。



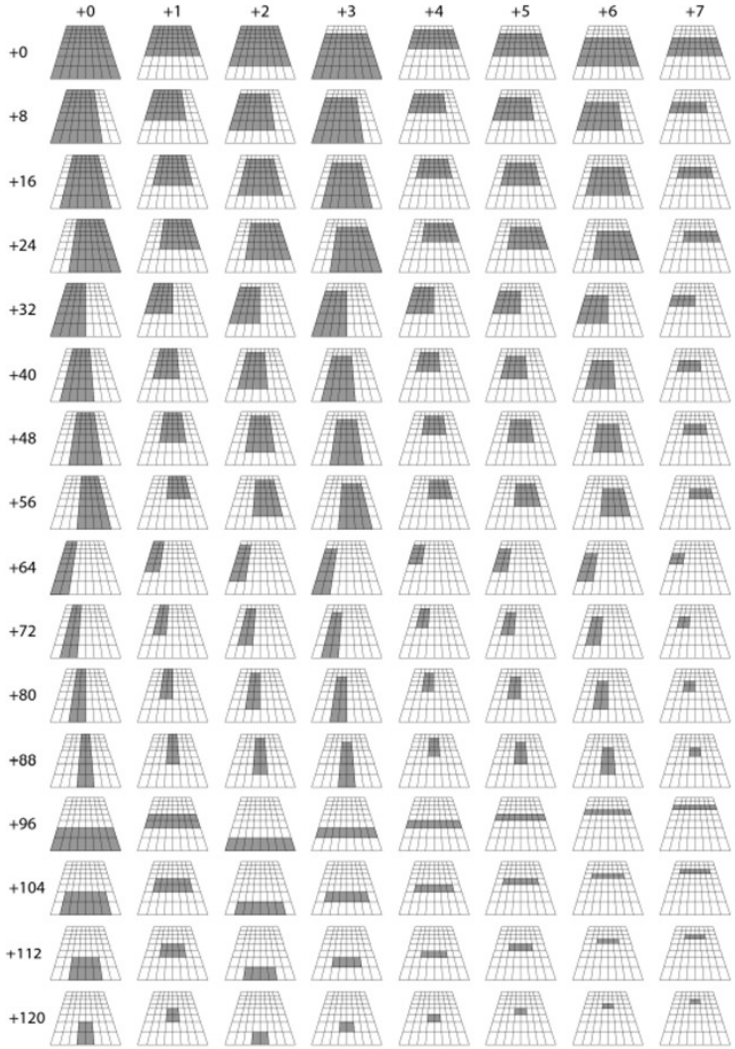
DS Camera

DS910B Series Technical Data

Data	DS910B
Measuring Range Z-axis	8 mm
Start of Measuring Range	52.5 mm
End of Measuring Range	60.5 mm
Line Length Midrange (X-axis)	10 mm
Linearity[1]	± 0.17 % FSO (3 σ)
Resolution X-axis	1280 points/profile
Light Source Laser	Semiconductor laser, approx. 405 nm, 10° aperture angle, Laser class 2M: laser power 7 mW, reduced 2 mW
Displays	1x state / 1x laser on
Electromagnetic Compatibility (EMC)	According to: EN 61326-1: 2006-10 DIN EN 55011: 2007-11 (Group 1, Class B) EN 61000-6-2: 2006-03

FSO = Full Scale Output | MMR = Midrange

DS Camera



DS Camera

Maximum Scan Rates for Measuring Fields

Maximum Acquiring Rate Depending on the Measuring Field Number								
	+0	+1	+2	+3	+4	+5	+6	+7
0	144Hz	192Hz	192Hz	192Hz	284Hz	284Hz	284Hz	552Hz
8	181Hz	239Hz	239Hz	239Hz	354Hz	354Hz	354Hz	680Hz
16	181Hz	239Hz	239Hz	239Hz	354Hz	354Hz	354Hz	680Hz
24	181Hz	239Hz	239Hz	239Hz	354Hz	354Hz	354Hz	680Hz
32	241Hz	318Hz	318Hz	318Hz	469Hz	469Hz	469Hz	892Hz
40	241Hz	318Hz	318Hz	318Hz	469Hz	469Hz	469Hz	892Hz
48	241Hz	318Hz	318Hz	318Hz	469Hz	469Hz	469Hz	892Hz
56	241Hz	318Hz	318Hz	318Hz	469Hz	469Hz	469Hz	892Hz
64	362Hz	476Hz	476Hz	476Hz	694Hz	694Hz	694Hz	1200Hz
72	362Hz	476Hz	476Hz	476Hz	694Hz	694Hz	694Hz	1200Hz
80	362Hz	476Hz	476Hz	476Hz	694Hz	694Hz	694Hz	1200Hz
88	362Hz	476Hz	476Hz	476Hz	694Hz	694Hz	694Hz	1200Hz
96	552Hz	552Hz	1030Hz	1030Hz	1030Hz	1030Hz	1030Hz	1030Hz
104	680Hz	680Hz	1200Hz	1200Hz	1200Hz	1200Hz	1200Hz	1200Hz
112	892Hz	892Hz	1200Hz	1200Hz	1200Hz	1200Hz	1200Hz	1200Hz
120	1200Hz	1200Hz	1200Hz	1200Hz	1200Hz	1200Hz	1200Hz	1200Hz

DS Camera

Recommended Shutter Times

The value of **exposure** (in milliseconds) should be set depending on the material that is being scanned. The recommendations are as follows:

Recommended shutter times (approximate values)	
Target material	**Shutter Time**
White paper/plastic	10 - 50µs
Colored plastic	50 - 100µs
Metallic surfaces	0.1 - 1ms
Black plastic/rubber	0.5 - 1ms

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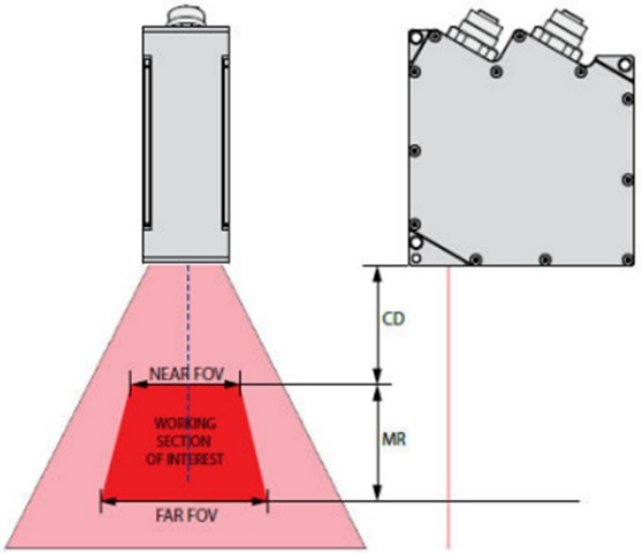
DS Camera

DS910B – Measurement Specifications

Specification	DS910B
Near Field of View	9.4 mm
Far Field of View	10.7 mm
Clearance Distance (CD)	53 mm
Measurement Range (MR)	8 mm
Resolution X	0.0073 mm - 0.0084 mm
Resolution Z	0.001 mm



Work Hard
Play Hard
Move Fast!